Course Overview

- Basic Structure of Computers
- Machine Instructions And Programs
- Control Unit
- Arithmetic And Logic Unit
- Memory System
- Input/Output Organization
- Pipelining

Functional Units

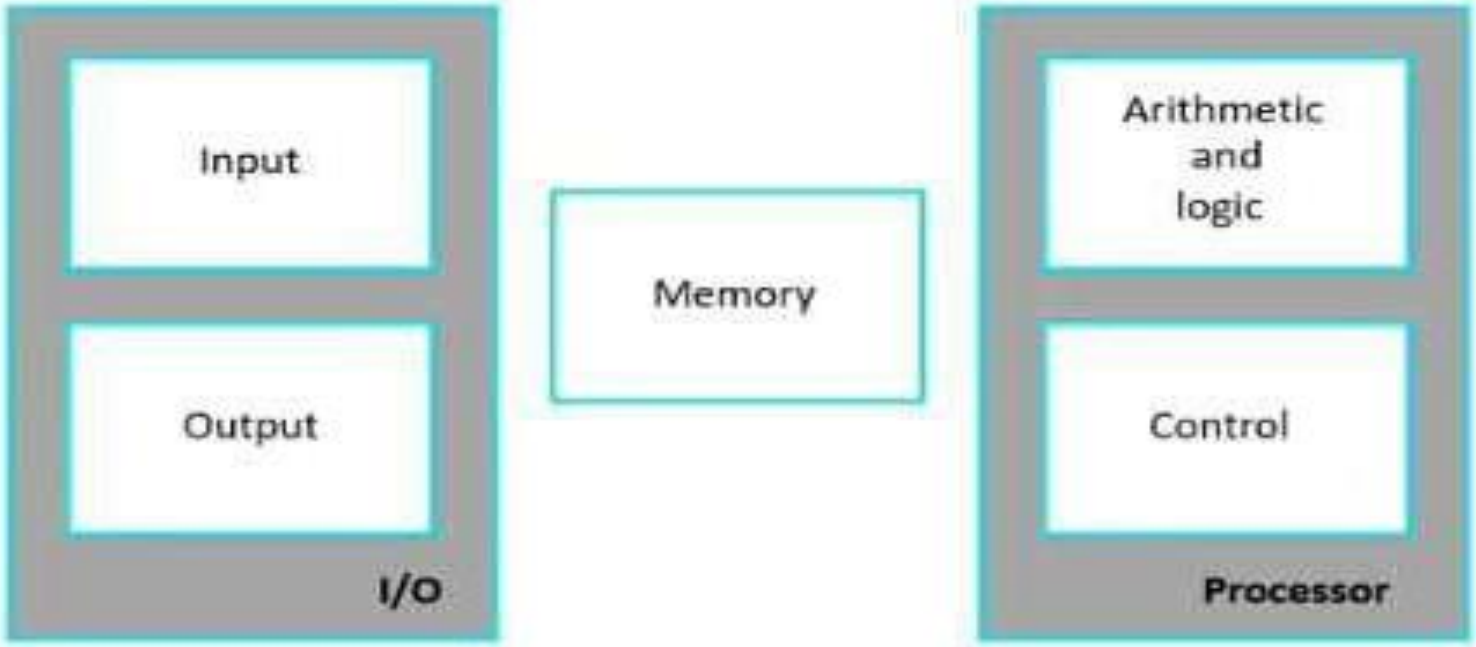


Figure 1.1. Basic functional units of a computer.

Information Handled by a Computer

- Instructions/machine instructions
  - Govern the transfer of information within a computer as well as between the computer and its I/O devices
  - Specify the arithmetic and logic operations to be performed
  - Source Program
- Data
  - Used as operands by the instructions
  - Source program can be considered as data to compilers
- Encoded in binary code – 0 and 1

Memory Unit

- Store programs and data
- Two classes of storage
  - Primary storage
    - Fast
    - Programs must be stored in memory while they are being executed
    - Large number of semiconductor storage cells
    - Processed in words
    - Need an **address** to fetch the memory word
    - RAM and memory access time
    - Memory hierarchy – cache, main memory
  - Secondary storage – larger and cheaper
    - Disks, CD-ROMs, Optical Disks

Arithmetic and Logic Unit (ALU)

- Most computer operations are executed in the ALU of the processor.
- Load the operands from memory – bring them to the processor – perform operation in ALU – store the result back to memory or retain in the processor.
- **Registers** to store values in the processor.
- ALU and control unit are much faster than other devices connected to a computer system.

Control Unit

- All computer operations are controlled by the control unit.
- The timing signals that govern the I/O transfers are also generated by the control unit.
- Control unit is usually distributed throughout the machine instead of standing alone.

Operations of a Computer

- Accept information in the form of programs and data through an input unit and store it in the memory
- Fetch the information stored in the memory, under program control, into an ALU, where the information is processed
- Output the processed information through an output unit
- Control all activities inside the machine through a control unit

Basic Operational Concepts

- Activity in a computer is governed by instructions.
- To perform a task, an appropriate program consisting of a list of instructions is stored in the memory.
- Individual instructions are brought from the memory into the processor, which executes the specified operations.
- Data to be used as operands are also stored in the memory.